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Assignment 10

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 51 = 3 + 48 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:
3 points: Correct answer with work shown
2 points: Incorrect answer but attempt shows some understanding (work shown)
1 point: Incorrect answer but an attempt was made (work shown)
0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]
(link to your live google doc)

https://docs.google.com/document/d/1dzAgyy7UhyrmR9CyIVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing

I. Inference for Categorical Data: Chi-Square (8 x 3 = 24 pts)

1) Are more babies born on a specific day of the week than others? To determine if the distribution of births across the week happened in different proportions than expected, a researcher took a random sample of 84 births in the year from a local hospital and recorded what day they were born on. The data is given in the following table:

Day of the week	Sun	Mon	Tue	Wed	Thu	Fri	Sat
# of births	6	7	9	14	19	17	12

Based on these data, is it reasonable to conclude that the proportion of births is not the same for all days of the week? Use $\alpha = 0.05$. [Use the 4 Step Process to Perform a proper Chi-Square Test]

Answer:

Step 1:

State:
H_0: The birthdays are evenly distributed among the days of the week.
H_a: The birthdays are not evenly distributed among the days of the week.
$\alpha = 0.05$

Step 2:

Plan: If the conditions are met, we will perform a chi-square test for goodness of fit.
Random: Random sample stated.
10%: Assume there are more than 840 births in the hospital on this week.
Large Counts: If birthdays are evenly distributed across the week then the expected counts are all 84(1/7) = 12. These counts are all at least 5.

Step 3:

Do: Calculate test statistic.

$X^2 = (6-12)^2/12 + (7-12)^2/12 + (9-12)^2/12 + (14-12)^2/12 + (19-12)^2/12 + (17-12)^2/12 + (12-12)^2/12 = 3.0 + 2.1 + 0.8 + 0.3 + 4.1 + 2.1 + 0 = 12.3$

df = 6

p-value is between 0.1 and 0.05.

Step 4:

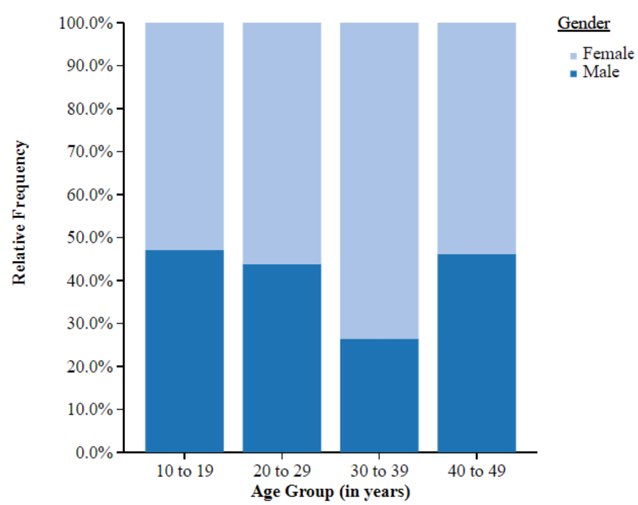
Conclude:

Because the p-value, 0.055, is greater than $\alpha = 0.05$ we fail to reject the null hypothesis.

We do not have convincing evidence that the births are not evenly distributed across the week days.

2) Dissociative identity disorder (DID) is a mental health condition where people diagnosed have two or more separate identities. These personalities control their behavior at different times. A random sample of 200 people currently diagnosed with DID in the US was taken and their ages at their diagnosis and their gender were recorded.

		Age Group (Years)				
		10 to 19	20 to 29	30 to 39	40 to 49	Total
Gender	Male	8	35	13	25	81
	Female	9	45	36	29	119
	Total	17	80	49	54	200



Do the data provide convincing statistical evidence of an association between age-group and gender in the diagnosis of DID? [Use the 4 Step Process to Perform a proper Chi-Square Test]

Answer:

Step 1:

State:
H_0: There is no association between age-group and gender in the diagnosis of DID.
H_a: There is an association between age-group and gender in the diagnosis of DID.
Significance level not stated, use $\alpha = 0.05$.

Step 2:

Plan: If the conditions are met, we will perform a chi-square test for goodness of fit.
Random: Random sample stated.
10%: Assume there are more than 2000 people with DID in the US.
Large Counts: If there is no association between age-group and gender in the diagnosis of DID then the expected counts are all at least 5. (6.9 is smallest)

Step 3:

Do: Calculate test statistic.
$X^2 = (8-6.9)^2/6.9 + (9-10.1)^2/10.1 + (35-32.4)^2/32.4 + (45-47.6)^2/47.6 + (13-19.8)^2/19.8 + (36-29.2)^2/29.2 + (25-21.9)^2/21.9 + (29-32.1)^2/32.1 = 0.2 + 0.1 + 0.2 + 0.1 + 2.4 + 1.6 + 0.4 + 0.3 = 5.4$
$df = (2 - 1)(4 - 1) = 3$
p-value is between 0.2 and 0.1.

Step 4:

Conclude:
Because the p-value, 0.15, is greater than $\alpha = 0.05$ we fail to reject the null hypothesis.

We do not have convincing evidence that there is an association between age-group and gender in the diagnosis of DID.

II. Inference for Quantitative Data: Slopes (8 x 3 = 24 pts)

1) A service center for kitchen appliances has hired a large group of new technicians. During the first 20 weeks of their employment, data were collected documenting the number of appliances serviced by a random sample of these technicians.

Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Number	12	21	18	13	28	20	29	35	26	20	36	40	28	52	32	32	55	56	49	60

Here is part of the R regression output for these data:

```
Predictor        Coef       StDev          T        P
Constant       10.679       3.554       3.00    0.008
Week           2.1353      0.2967       7.20    0.000

S = 7.651        R-Sq = 74.2%     R-Sq(adj) = 72.8%
```

(a) Interpret the slope of the regression in the context of the problem.

Answer:

The slope represents how many additional appliances are serviced after each additional week of employment.

(b) What is the equation for the LSRL line?

Answer:

y-hat = 10.679 + 2.1353x

(c) Using the R output, determine a 99% confidence interval for the true slope of the regression line. Just carry out the calculation

Answer:

$\beta = b \pm t^* SE_b$

df = n -2 = 20 - 2 = 18

t* = 2.552

SE = S/S_x*sqrt(n-1) = 7.65/(5.25*sqrt(19)) = 0.3350

$\beta = 2.1353 \pm 2.552 * 0.3350 = 2.1353 \pm 0.854$

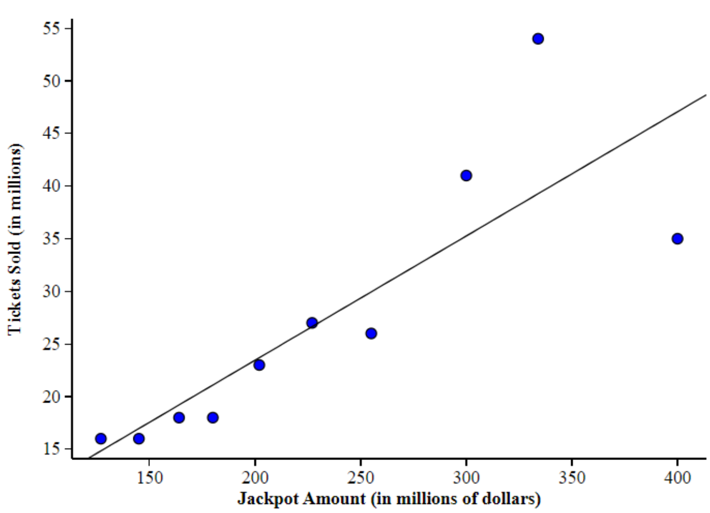
The 99% confidence interval for the true slope is (1.2813, 2.9893).

(d) Interpret the meaning of the confidence interval in the context of the problem. What does this interval suggest about the employment of the new technicians?

Answer:

This confidence interval means that we are 99% confident that the interval captures the slope of the population regression, which means new technicians will service between 1.2813 and 2.9893 additional appliances for each additional week of employment. This interval means that there is an association between the number of appliances and the number of weeks employed.

2) The largest winning jackpot in Mega Millions history was $1.537 billion, for the October 23, 2018, drawing, in which a single winning jackpot ticket was sold in South Carolina. Ticket sales leading up to the Jackpot being won were unprecedented. A statistician was interested in determining if the number of tickets sold for a Mega Millions drawing depended on how large the Jackpot was. They randomly selected 10 Mega Millions drawings in history and recorded how many ticket sales there were for that drawing. The data below shows the result of the investigation.



r = 0.836

$s_x = 88.57$

Regression Analysis

Equation	n	s
$\hat{y} = -0.147 + 0.118$	10	7.288

T-Test for Slope

$H_0: \beta = 0$

$H_a: \beta > 0$

(a) What is the standard error of the slope of the regression line? Interpret it in the context of the problem.

Answer:

SE = S/(Sx*sqrt(n-1)) = 7.288/(88.57*sqrt(9)) = 0.0274

The standard error represents the amount of discrepancy in a sample estimate compared to the true value in the population. In this case, the standard error represents how close the predicted number of tickets sold is to the real value.

(b) What do you know (given the information about) or what would you need to know in order to perform a T-Test for the slope of the regression line?

Answer:

We know there is a linear relationship from the scatterplot.
We can assume there are more than 100 Mega Millions jackpots.
We need to know if the histogram of residuals is approximately normal.
We would also need to know if the residual plot has random scatter.
We know that the sample is random.

(c) Assuming that the conditions needed for doing inference for regression are present, what is the correct t-test statistic for the listed hypothesis?

Answer:

$t = (b - \beta_0)/SE = 0.118/0.0274 = 4.3066$
$b = 0.118$
$\beta_0 = 0$
$SE = 0.0274$

(d) Find and interpret the p-value with the test statistic you calculated in part (c). What is the conclusion to this test in the context of the problem? Use a significance level of 5%.

Answer:

p-value = tcdf(4.3066,1E99,8) = 0.00129

Because the p-value is less than $\alpha = 0.05$ we reject the null hypothesis and conclude there is an association between jackpot value and tickets sold.

THE END